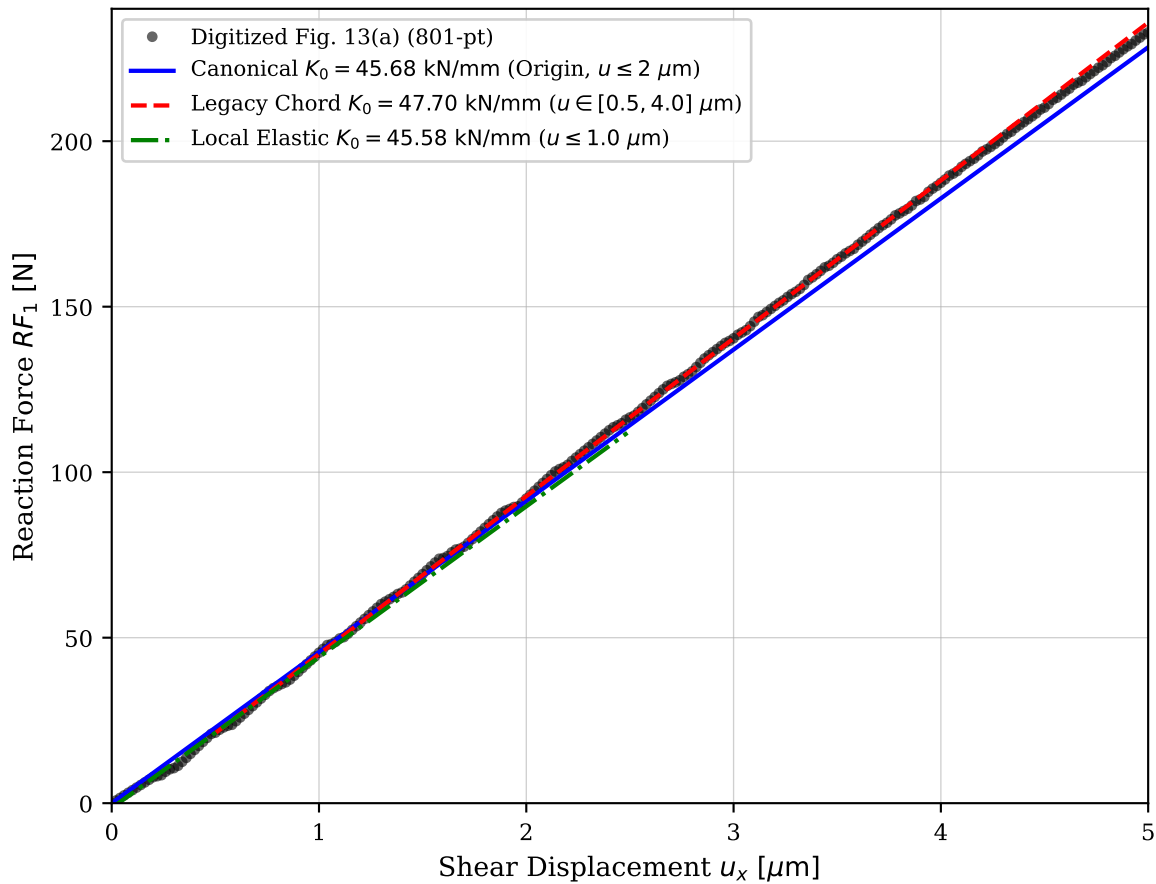
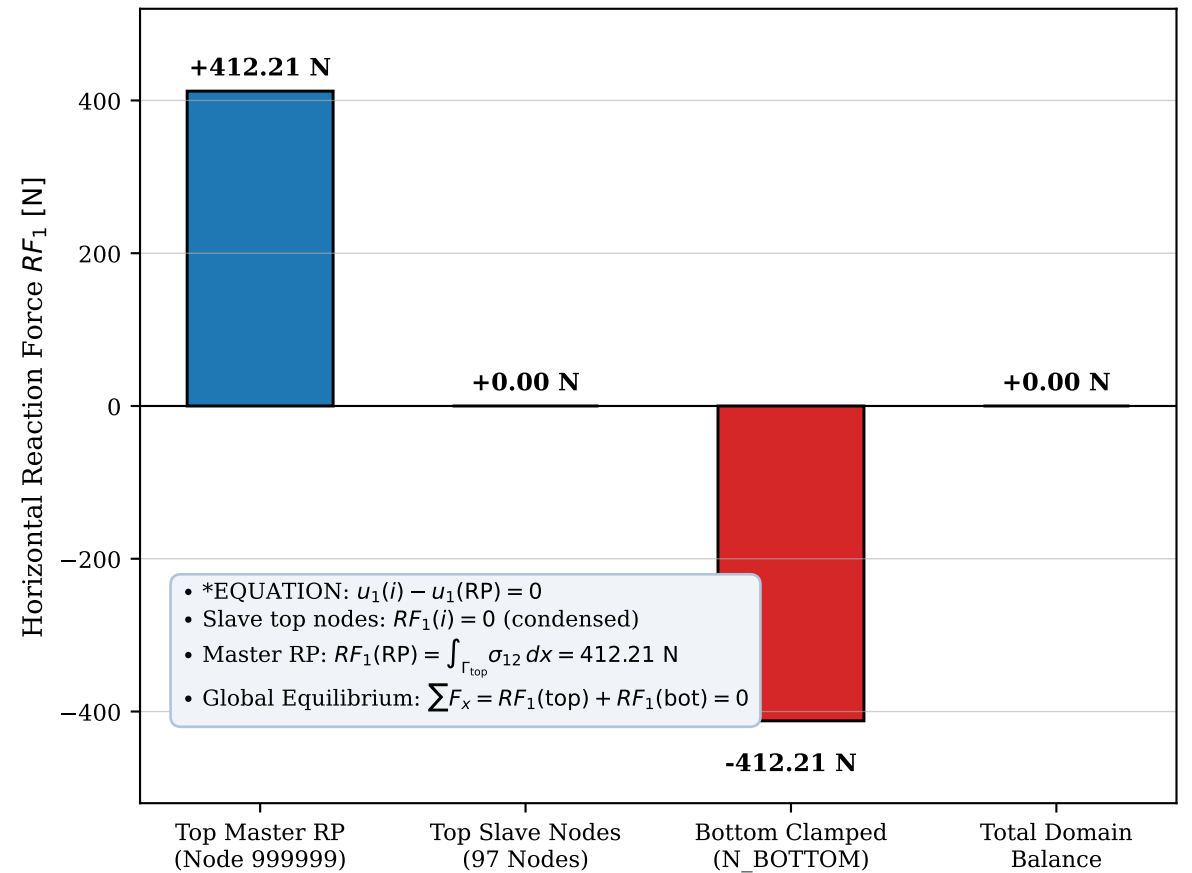


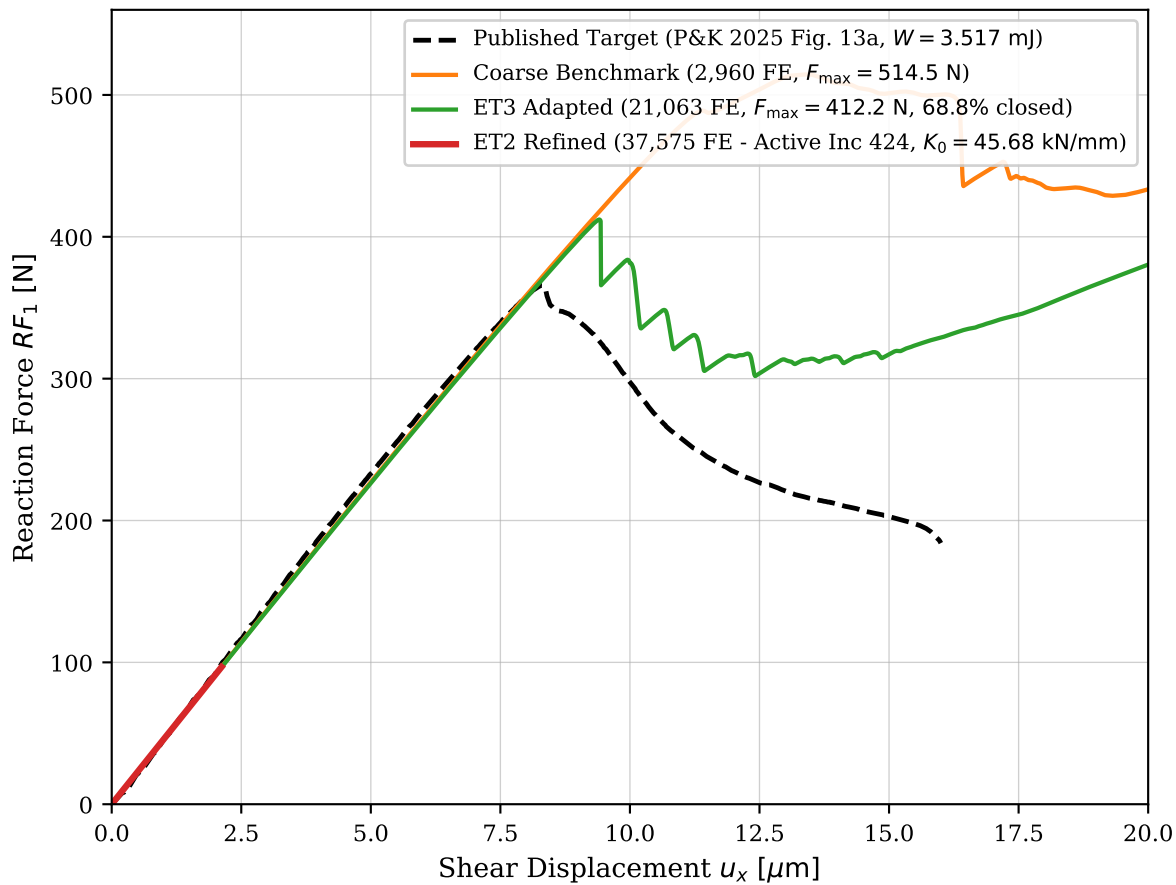
(a) Reference Initial Stiffness Reconciliation (Pandey & Kumar Fig. 13a)



(b) Global Boundary Equilibrium & *EQUATION MPC Verification



(c) Multi-Discretization Mechanical Comparison & Active ET2 Tracking

(d) Crack Deceleration ($15 \times$) & Boundary Ligament Approaching Base